

YEAR 12 ASSESSMENT TEST

MATHEMATICS 4 UNIT

March 1998

Time Allowed: 1.5 hour

Question 1

- (a) Using DeMoivre's theorem, or otherwise, find the square roots of $\sqrt{3} + i$
- (b) Sketch on an Argand diagram, the locus of z defined by:
- i) $\text{Arg}\left(\frac{z-i}{z+1}\right) = 0$ ii) $|z - (2+3i)| = 25$
- (c) A polynomial $P(x)$ gives remainders of 2 and 6 when divided by $x-1$ and $x+1$ respectively. What must the remainder be when $P(x)$ is divided by x^2-1 ?
- (d) Sketch the ellipse $x^2 + 4y^2 = 16$, showing the foci and the directrices.
- (e) State the domain and range of $y = 2 \sin^{-1}\left(\frac{3x-1}{2}\right)$ and make a neat sketch of the curve.
- What is the slope of y when $x = \frac{1}{3}$?

Question 2 (Start a new page)

- (a) Given that a monic polynomial $Q(x)$ with rational coefficients has one root $1+2i$ and that 3 is a root of multiplicity 2, write the equation of the lowest possible degree of $Q(x)$.
- (b) Find the equation of the tangent to the ellipse $4x^2 + 9y^2 = 36$ at the point $\left(1, \frac{4\sqrt{2}}{3}\right)$.
- Show that this tangent passes through the point $\left(6, \frac{1}{\sqrt{2}}\right)$. Hence find the points on the ellipse where the chord of contact from the point $\left(6, \frac{1}{\sqrt{2}}\right)$ intersects the ellipse.

i) Factorise $z^2 - 1$ over the complex field

ii) Show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = \frac{-1}{2}$

iii) If ω is one of the complex fifth roots of unity, show that $\omega + \omega^4$ and $\omega^2 + \omega^3$ are the roots of the quadratic equation $x^2 + x - 1 = 0$

$$\omega + \omega^4 \quad \& \quad \omega^2 + \omega^3$$

Question 3 (Start a new page)

- (a) Find the equation of the tangent to the rectangular hyperbola $xy = c^2$ at the point $\left(ct, \frac{c}{t}\right)$. Also find where this tangent meets the x and y axis.
- (b) Given that the polynomial $P(x) = x^5 - 6x^4 + 6x^3 + 20x^2 - 39x + 18$ has two roots of multiplicity two, solve $P(x) = 0$
- (c) i) State the domain of $y = (x-2)\sqrt{x-1}$
- ii) Differentiate $y = (x-2)^2(x-1)$ and evaluate
- (a) as $x \rightarrow 1$
- (b) as $x \rightarrow -\infty$
- (c) when $x = 2$
- iii) Hence draw a neat sketch of the graph $y^2 = (x-2)^2(x-1)$.

Question 4 (Start a new page)

(a) Resolve $\frac{x}{(x-3)(x+1)}$ into partial fractions, hence evaluate $\int_4^5 \frac{2x}{(x-3)(x+1)} dx$

(b) i) Show that the condition for the line $y = mx + c$ to touch the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $c^2 = a^2 m^2 - b^2$

ii) If P is the foot of the perpendicular from a focus S of the hyperbola to a variable tangent with slope m , by eliminating m from the equation of the tangent and perpendicular, find the locus of P .

(c) $P(3 \sec \theta, 2 \tan \theta)$, and $Q(3 \sec \theta, -2 \tan \theta)$ are two points on a hyperbola of centre O .

i) Find the equation of the hyperbola. Make a sketch of the hyperbola, showing the foci, directrices and asymptotes.

ii) Plot the points P, Q for $\theta = \frac{\pi}{3}$

iii) If the normal to the hyperbola at P meets OQ in R , show that the locus of R is a hyperbola of centre O .

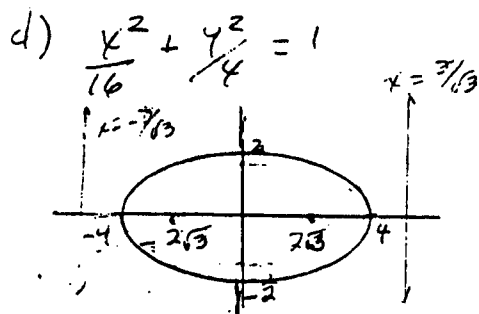
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Solutions

Q1 a) Let $z^2 = r^2(\cos \theta)^2 = \sqrt{3} + i$
 $r^2 = (\sqrt{3})^2 + 1 = 2$ so $r = \sqrt{2}$
 $(\cos \theta)^2 = \cos 2\theta$
 $\cos 2\theta = \sqrt{3}/2$ so $2\theta = \frac{\pi}{2}$
 $\theta = \frac{\pi}{4}, -\frac{\pi}{4}$ so 2 roots
are $\sqrt{2} \cos(\frac{\pi}{4})$ and $\sqrt{2} \cos(-\frac{\pi}{4})$

c) $P(x) = (x^2 - 1)Q(x) + 4x + 6$
 $P(1) = a + 6 = 2$
 $P(-1) = -a + 6 = 6$
 $2b = 8$
 $b = 4$
 $a = -2$

so remainder is $-2x + 4$

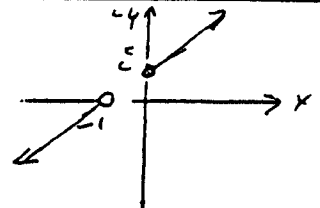


foci $(\pm ae, 0)$ $ae = \sqrt{a^2 - b^2}$
 $= 2\sqrt{3}$

directrices $y = \pm \frac{a}{e}$
 $= \pm \frac{4}{\sqrt{3}}$

$\frac{dy}{dx} = \frac{2 \times \frac{3}{2}}{1 - (\frac{3x-1}{2})^2}$
slope = 3 when $x = \frac{1}{3}$

b) c)



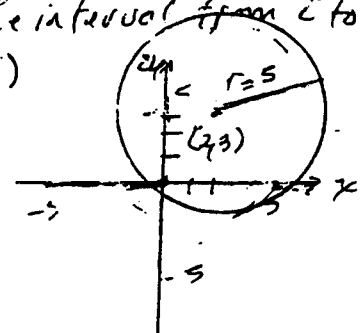
$\arg(\frac{z-i}{2-i}) = 0$

$\arg(z-i) = \arg(2-i)$

$\arg(z-i) = \arg(2-i)$

Line: $y = x + 1$ but not the interval from i to -1

c)



$|z - (2+3i)| = 5$

$(x-2)^2 + (y-3)^2 = 25$

$(x-2)^2 + (y-3)^2 = 25$

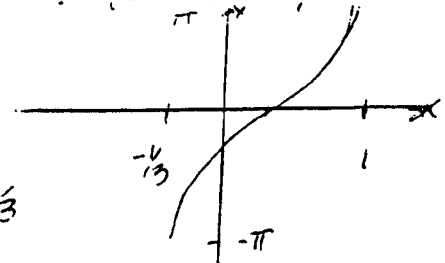
a circle centre $(2, 3)$ radius 5

e) $D: -1 \leq \frac{3x-1}{2} \leq 1$

$\therefore D: -\frac{1}{3} \leq x \leq 1$

$R: -\frac{\pi}{2} \leq \arctan(\frac{3x-1}{2}) \leq \frac{\pi}{2}$

$\therefore R: -\pi \leq y \leq \pi$



Q2) a) $1-2i$ is also a root since coefficients are rational. so

$$\text{equation } P(x) = (x-3)^2(x-1+2i)(x-1-2i)$$

$$= x^4 - 8x^3 + 26x^2 - 48x + 45$$

$$\text{or } P(x) = x^4 - 8x^3 + 26x^2 - 48x + 45$$

$$\text{where } \alpha, \beta = 3 \quad \gamma = 1+2i \quad \delta = 1-2i$$

$$b) \frac{x^2}{9} + \frac{y^2}{4} = 1 \quad \text{at } (1, \frac{4\sqrt{2}}{3})$$

$$\therefore \text{tangent is } \frac{x}{9} + \frac{\sqrt{2}}{3}y = 1 \quad \text{at } (1, \frac{4\sqrt{2}}{3})$$

$$\text{now } \frac{x}{9} + \frac{\sqrt{2}}{3}y = \frac{2}{3} + \frac{1}{3} \quad \text{for } (6, \frac{1}{\sqrt{2}})$$

$$= 1$$

$\therefore$ tangent contains $(6, \frac{1}{\sqrt{2}})$

$$\text{Chord of Contact is } \frac{6x}{9} + \frac{y}{4\sqrt{2}} = 1$$

$$\text{so } y = 4\sqrt{2}(1 - \frac{2}{3}x)$$

for pts of intersection

$$\frac{x^2}{9} + \frac{[4\sqrt{2}(1 - \frac{2}{3}x)]^2}{4} = 1$$

$$x^2 + 72(1 - \frac{2}{3}x)^2 = 9 \quad \text{on simplifying}$$

coeff of x^2 is 33, constant is 63

$$\text{so } a \cdot b = \frac{21}{11} \quad x = 1, \text{ so } z = \frac{21}{11}$$

$$\text{so pts are } (1, \frac{4\sqrt{2}}{3}) \text{ and } (\frac{21}{11}, -\frac{12\sqrt{2}}{11})$$

Q3) a) $x \frac{dy}{dx} + y \frac{dx}{dx} = 0 \quad \frac{dy}{dx} = -\frac{y}{x}$

$$= -\frac{y}{x} \quad \text{when } x = x \quad \frac{dy}{dx} = -\frac{y}{x}$$

$$\therefore T: y - y_1 = -\frac{y_1}{x_1}(x - x_1)$$

$$\text{If } x = 2 \text{ at } x = 0 \text{ is } y = 0$$

$$x_1 = 0, y_1 = 0 \quad \text{and } y_1 = x_1, y = 0 \quad (\frac{0}{2}, \frac{2}{2})$$

$$b) P(x) = 5x^4 - 24x^3 + 18x^2 + 40x - 39$$

$$P'(1) = 5 - 24 + 18 + 40 - 39 = 0$$

also $P(1) = 0$ so 1 is a double root

try factors of 18 say $P'(3) = 0$ or $P(3) = 0$

now product of roots = -18 so

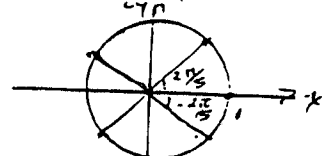
$P(2) = 0$ also note: only 5 solutions are possible: 1, 1, 3, 3, 2

c) let $z = r \cos \theta$

$$\text{then } z^5 = \cos 5\theta = 1$$

so roots are $\cos \frac{2n\pi}{5}$

for $n = 0, \pm 1, \pm 2$



5 equally spaced roots on circle $x^2 + y^2 = 1$

$$i) \text{ factors are } (z-1)(z-\omega)(z-\omega^2)(z-\omega^3)(z-\omega^4)$$

$$ii) \text{ sum of roots } z^5 - 1 =$$

$$= 0$$

$$\text{so } 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$$

$$\omega + \omega^4 + \omega^2 + \omega^3 = -1$$

$$\omega + \omega^4 = 2\cos \frac{4\pi}{5}$$

$$\omega^2 + \omega^3 = 2\cos \frac{2\pi}{5}$$

$$\text{so } 1 + 2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5} = 0$$

$$i) \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

$$ii) 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$$

$$(\omega + \omega^4) + (\omega^2 + \omega^3) = -1$$

$$\text{also } (\omega + \omega^4)(\omega^2 + \omega^3) =$$

$$\omega^3 + \omega^4 + \omega^6 + \omega^7 =$$

$$\omega^3 + \omega^4 + (\omega^6 + \omega^7) \quad \text{since}$$

$$\omega^6 = \omega^2 \quad \omega^7 = \omega$$

$$\therefore \text{ product} = -1$$

$$\text{so quadratic is } x^2 + x + 1 =$$

Q3) i) domain $x \geq 1$

ii) $y^2 = (x-2)^2(x-1)$

$2y dy = [2(x-2)(x-1) + (x-2)^2] dx$

$\frac{dy}{dx} = \frac{(x-2)(3x-4)}{2y}$

$= (x-2)(3x-4)$

$\pm 2(x-2)\sqrt{x-1}$

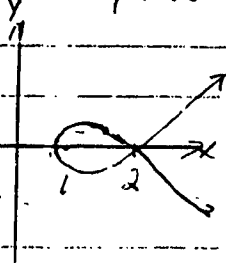
$= \pm (3x-4)$

$2\sqrt{x-1} \quad x \neq 1, 2$

a) as $x \rightarrow 1 \quad \frac{dy}{dx} \rightarrow \pm \infty$

b) as $x \rightarrow +\infty \quad \frac{dy}{dx} \rightarrow \pm \infty$

c) when $x=2$ undetermined
but ± 1 in simplified $\frac{dy}{dx}$



Q4) a) $\frac{x}{(x-3)(x+1)} = \frac{a}{x-3} + \frac{b}{x+1}$

$x = a(x+1) + b(x-3)$

if $x = -1 \quad -1 = -4b$

$\frac{1}{4} = b$

if $x = 3 \quad 3 = 4a$

$\frac{3}{4} = a$

$\int \frac{x}{(x-3)(x+1)} dx = \int \left[\frac{3}{4(x-3)} + \frac{1}{4(x+1)} \right] dx$

$= \frac{3}{4} \int \frac{1}{x-3} dx + \frac{1}{4} \int \frac{1}{x+1} dx$

$= \frac{3}{4} (\ln 2 - \ln 1) + \frac{1}{4} (\ln 4 - \ln 5)$

$= \frac{3}{4} \ln 2 + \frac{1}{4} \ln \frac{4}{5}$

b) i) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad y = mx + c$

$\frac{x^2}{a^2} - \frac{(mx+c)^2}{b^2} = 1$

$(b^2 - a^2m^2)x^2 - 2amcx - a^2(b^2/a^2 + c^2/b^2) = 0$

letting $\Delta = 0$

$4a^2mc^2 - 4(b^2 - a^2m^2)(a^2(b^2/a^2 + c^2/b^2)) = 0$

$4a^2b^2(-b^2 - c^2 + a^2m^2) = 0$

$c^2 = a^2m^2 - b^2$

ii) perpendicular at focus

$y - 0 = -\frac{1}{m}(x - ac)$

$x + my = ac$

tangent from (c)

$y = mx + \sqrt{a^2m^2 - b^2}$

squaring both

$(x + my)^2 = a^2c^2$

$(y - mx)^2 = a^2m^2 - b^2$

Adding $(1+m^2)y^2 + (1+m^2)x^2 = a^2m^2 - b^2 + a^2c^2$

subst $b^2 = a^2(c^2 - 1)$

RHS $a^2(1+m^2)$

$\therefore x^2 + y^2 = a^2$ (divide by $1+m^2$)
circle radius a

and centre $(0,0)$

(4) (i) $P = (3 \sec \theta, 2 \tan \theta)$

$$x = 3 \sec \theta \quad y = 2 \tan \theta$$

$$\sec \theta = \frac{x}{3} \quad \tan \theta = \frac{y}{2}$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

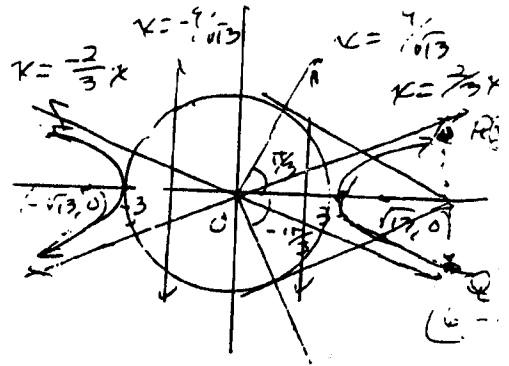
$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$ae = \sqrt{a^2 + b^2} \quad (\text{foci})$$

$$= \sqrt{9 + 4}$$

$$\frac{a}{e} = \frac{3}{\sqrt{13}} \quad (\text{directrices})$$

$$\pm \frac{3}{a} x = \pm \frac{2}{b} y \quad (\text{asymptotes})$$



(ii) On diagram

(iii) Normal: $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$

Equation of P $y = -\frac{2 \tan \theta}{3 \sec \theta} x$

Subst: $\frac{ax}{\sec \theta} + \frac{b(-2x)}{3 \sec \theta} = a^2 + b^2$

Solving $\sec \theta = \frac{(3a - 2b)x}{3(a^2 + b^2)}$

Subst $x = \frac{3 \sec \theta}{-2 \tan \theta}$ into Normal

$$\frac{3ay}{-2 \tan \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$

Solving $\tan \theta = \frac{(3a - 2b)y}{-2(a^2 + b^2)}$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\frac{(3a - 2b)^2 x^2}{9(a^2 + b^2)^2} - \frac{(3a - 2b)^2 y^2}{4(a^2 + b^2)^2} = 1$$

which is a hyperbola centre (0,0).